IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 15 September 2023, accepted 22 October 2023, date of publication 1 November 2023, date of current version 6 November 2023.

Digital Object Identifier 10.1109/ACCESS.2023.3329257

RESEARCH ARTICLE

Designing and Evaluating User Experience of an AI-Based Defense System

SUNYOUNG PARK[1], HYUN K. KIM[1,2], JAEHYUN PARK[3], AND YURYEON LEE[1]

[1]Department of Artificial Intelligence Convergence, Kwangwoon University, Seoul 01897, South Korea
[2]IAE Laboratory, School of Information Convergence, Kwangwoon University, Seoul 01897, South Korea
[3]Department of Industrial and Management Engineering, Incheon National University, Incheon 22012, South Korea

Corresponding author: Hyun K. Kim (hyunkkim@kw.ac.kr)

This work was supported by the National Research Foundation of Korea (NRF) funded by the Korean Government (MSIT) under Grant NRF-2021R1F1A1063155.

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Board of Kwangwoon University under Application No. 7001546-20210701-HR(SB)-007-04.

ABSTRACT In recent years, artificial intelligence (AI) has been applied in various fields, with rapid expansion of the scope of AI-human interactions. However, most AI technologies continue to exhibit black-box characteristics, i.e., their decisions and actions are not explainable, degrading user experience (UX). Recently, research on explainable AI (XAI), focusing on both AI performance and UX, has garnered significant attention. However, development of generalizable UX evaluation tools for AI and improvement of AI UX have not been investigated adequately. In this study, a UX evaluation tool is developed for AI based on a systematic literature review and verified using exploratory and confirmatory factor analyses. Subsequently, based on identified AI UX factors, UX of an AI defense system is upgraded. Based on user evaluation, there were significant improvements in eight factors (satisfaction, safety, controllability, trust, causality, efficiency, accountability, and explainability) with the exception of fairness. The proposed evaluation tool is expected to serve as a cornerstone for future evaluation of AI UX advancement.

INDEX TERMS Artificial intelligence (AI), AI system design, AI UX, AI UX measurement, user experience (UX).

I. INTRODUCTION

Artificial intelligence (AI) has been applied in various fields, e.g., transportation, domestic/service robotics, medical care, education, public safety, and security, to improve the general standard of life [1]. McCarthy et al. [2] introduced AI (Schuett, 2019). According to McCarthy [3], intelligence refers to the computational aspect of the ability to achieve goals. AI has also been defined in terms of passing the Turing test, which determines whether a computer can think like a human [4]. According to Russell [5], an intelligent agent acts by receiving perceptions from the environment, and intelligence refers to the ability to select expected actions to optimize performance. Thus, although there is no general agreed-upon definition for AI, the aforementioned representative definitions demonstrate that AI is usually defined in terms of the ability of machines to imitate intelligent human behavior. AI-based technology may be of various types, e.g., image processing networks, computer vision architectures, artificial neural networks, machine learning-based techniques, convolutional neural networks, and deep learning-based methods [6]. These technologies are opaque and non-intuitive, i.e., their decisions and actions are not explainable [7]. This reduces the confidence of human users in results obtained using such systems, hindering the adoption of AI systems [8]. To combat this issue, user experience (UX) of AI systems has become an important research topic. UX represents the overall experience of users

The associate editor coordinating the review of this manuscript and approving it for publication was Giuseppe Desolda.

during their interaction with particular products or services [9], [10], [11]. In this context, active research is being conducted on explainable AI (XAI), human-AI collaboration, and human-AI designing AI.

Previous studies on XAI can be broadly classified into two categories. Those of the first type investigate the effects of certain factors on AI UX based on usability tests. Kulesza et al. [12] explained machine learning-based predictions to users and proposed a prototype that enables users to interact with the system. They reported a 52 % improvement in user comprehension compared to existing systems when descriptive debugging and more efficient error correction were provided. Park et al. [13] investigated the adequacy and availability of prior explanations of fake AI prototypes to users, as well as the effect of system performance on UX. According to research results, AI performance had a significant impact on users' trust, satisfaction, and emotions and higher AI performance led to a more positive UX. They concluded that prior explanations exert little influence on UX in the case of high-performance models but reported that trust and satisfaction was improved significantly in the case of low-performance models when the correct explanation was provided. Cheng et al. [14] investigated the effect of explanations on UX of 199 subjects in the context of decision-making algorithms used for university admission processes—interactive explanations and white-box explanations were reported to be effective in improving users' comprehension, even though they made the method more time-consuming. Bitkina et al. [15] evaluated perceived trust using Google's AI tool, AutoDraw. Their main results confirmed that users' perceived trust in new AI products can be increased via good product performance and successful implementation of tasks. In a follow-up study, Bitkina et al. [16] measured stress level indicators during task performance using English proofreading software and AutoDraw and determined and classified stress using a logistic regression model. The accuracies of the two stress models were approximately 70 %, and a dependence was observed between the electrodermal activity signal features, stress, and AI system trustworthiness. Reeder et al. [64] investigated how an individual's gender and educational background impact trust and comprehension of XAI in recommendation engines. Since both gender and educational background play significant roles in determining trust and comprehension of the system, they confirmed the importance of considering the characteristics of the study participants. Aechtner et al. [65] investigated user perceptions of local and global explanations using three different XAI techniques. The key findings revealed that explanations enhance user trust in the system, and novice AI users prefer local explanations over global ones.

By contrast, studies of the second type define factors that affect UX design and investigate dependencies among them. Wang et al. [17] proposed a conceptual framework for constructing human-centered XAI in medical decision-making processes based on theories derived from philosophy and psychology. Amershi et al. [18] proposed 18 design guidelines that are generally applied in human-AI interaction and classified them in terms of the interaction period (initially, during interaction, when wrong, or over time). Subsequently, the results were verified by practitioners who tested the guidelines for 20 popular AI-infused products. Shin [19] evaluated the effects of fairness, social responsibility, transparency, and explainability of AI systems on UX using a news-recommendation algorithm and proposed a conceptual model, confirming that transparency, causality, reliability, and social responsibility influence each other. Kieslich et al. [20] compared the public perception of the importance of each factor involved in the principles of ethical AI design using survey data obtained from 1204 participants residing in Germany. The study concluded that, on average, accountability was perceived as a more important ethical principle than fairness, security, privacy, accuracy, explainability, and machine autonomy. Zheng et al. [21] collected 1,302 documents related to conversational human-AI interactions from the Association for Computing Machinery digital library and analyzed them using thematic analysis and topic modeling based on term frequency-inverse document frequency scores. They concluded that social boundaries, such as privacy, disclosure, and identification, are very important in the design of ethical polyadic conversational agents. Hoffman et al. [66] have synthesized key concepts for measuring the quality and reliability of AI explanations. They propose a multifaceted approach to measuring aspects such as a priori goodness, satisfaction, mental model, curiosity, trust and reliance, and human-AI performance. Bach et al. [67] conducted a comprehensive review of 23 empirical studies to provide an overview of user trust definitions, influencing factors, and measurement methods concerning AI-enabled systems.

The aforementioned review yields two insights. First, several evaluation metrics have been developed targeting a wide range of AI and UX, focusing on specific situations, systems with specific characteristics, or factors considered to be important by researchers. This can be attributed to the unclear [22], ambiguous [23], and subjective definition of UX owing to the wide range of systems and services to which AI is applied. In this context, this study develops a generalizable evaluation index that can be used to evaluate all interactions of AI (including systems and services) with end users. Second, application of the established AI UX principle has not been investigated adequately. In this study, the AI UX principle is applied to the upgrading process, and an additional process is proposed to compensate for the insufficiency of only utilizing the principle.

The remainder of this manuscript is structured as follows. A tool is proposed and validated for AI UX evaluation in Section II, and the process of upgrading the UX of AI systems is described in Section III. The systems before and after applying these upgrades are compared in terms of UX in Section IV. Finally, future research directions are discussed in Section V, and our conclusions are summarized in Section VI.

II. DEVELOPMENT AND VALIDATION OF AI UX EVALUATION QUESTIONNAIRE

A. AI UX EVALUATION QUESTIONNAIRE

A three-step process is employed to develop initial questionnaire items (Figure 1). First, papers containing certain keywords, e.g., AI usability, AI system usability, AI trust, explainable AI, XAI, AI confidence, and interpretable AI, are collected to define AI UX elements. As AI is an abbreviation, a keyword search is performed by substituting it with Artificial Intelligence. In aggregate, 120 research papers are collected—49 from journals, 52 from conferences, and 19 from other sources. In the second step, high/medium/low filtering is performed on the selected literature by three researchers in terms of their degrees of relevance to AI UX—45 papers are deemed to exhibit high relevance; 34, medium relevance; and 41, low relevance. In the third step, concepts of elements are defined based on the selected literature sorted in order of degree of relevance. In aggregate, 12 AI UX components are defined as initial items—causality, fairness, accountability, explainability, transparency, performance, trust, satisfaction, safety, ease of use, efficiency, and controllability. These concepts are subsequently redefined based on the definitions and explanations of the terms in the literature, and survey items are created based on important concepts related to each term (Figure 2).

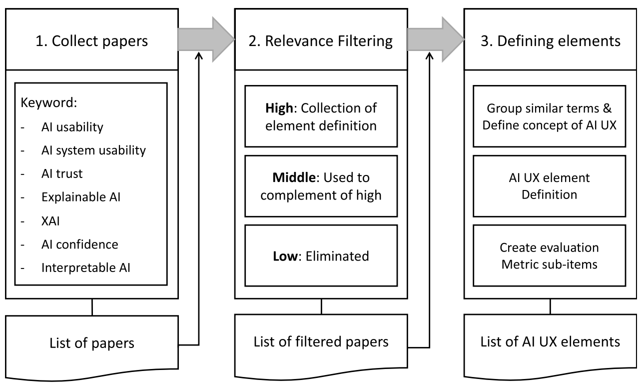


FIGURE 1. Development process of AI UX evaluation questionnaire.

B. USER SURVEY ON AI SPEAKERS

To verify the adequacy of the initial AI UX elements, an online survey is conducted on users of AI speakers using Google Forms. The questionnaire items include AI speaker usage behavior (e.g., frequency of use and service used) and AI speaker usage experience (items described in Section II-A). The results are filtered to remove ineligible participants using the following filtering criteria: 1) entry of inappropriate score, 2) same scores being recorded in more than 75 % of the subjective evaluation, 3) same scores being recorded 10 times successively in the subjective evaluation, and 4) less than three types of scores being recorded in the subjective evaluation. In aggregate, 134 people in their 20s are recruited to use AI speakers. 29 respondents are removed owing to overlapping participation or insincere answers, and data from the remaining 105 participants (65 men and 40 women) are finally used. The observed distribution of service usage is as follows: Siri (41.39 %), Bixby (28.27 %), Giga Gini (22.21 %), and Google Assistant (9.8 %) (Figure 3(a)). 49 participants (47 %) use the service at least once a day, 20 (19 %) use it at least three times a week, 16 (15 %) use it at least once a week, and 20 (19 %) use it less than once a week (Figure 3(b)).

C. FACTOR ANALYSIS

Cronbach's alpha and exploratory factor analyses are performed to confirm the reliability and validity of the items. The internal consistencies of the questionnaire items are evaluated as follows: satisfaction: 0.923; safety: 0.918; controllability: 0.876; trust: 0.925; causality: 0.845; fairness: 0.821; efficiency: 0.803; accountability: 0.727; and explainability: 0.799. The results of the exploratory factor analysis are presented in Table 1. The initially developed questionnaire consists of 12 factors and 45 detailed items; however, after performing exploratory factor analysis, satisfaction and performance are combined into a single factor, and two other factors are deleted. In addition, 22 sub-items are deleted, finally yielding 9 factors and 23 items (Table 1). Cronbach's alpha (α) analysis is repeatedly performed to remove items hindering internal consistency, and a backward-elimination process is executed to select appropriate questionnaire items. The following elimination criteria are used: (1) when the factor loading is less than 0.55, and (2) when one factor consists of fewer than two items. Principal component analysis is used as the factorization method and varimax rotation is used as the factor rotation method. Kaiser-Meyer-Olkin (KMO) and Bartlett's sphericity test are performed to verify the suitability of the dataset. In this study, the KMO value is observed to be 0.876, and Bartlett's significance probability is less than 0.001. To verify the questionnaire items, confirmatory factor analysis is employed. The measured values are Goodness of Fit Index (GFI) = 0.862, Root Mean Square Error of Approximation (RMSEA) = 0.057, Normed Fit Index (NFI) = 0.891, Comparative Fit Index (CFI) = 0.961, ratio of chi-square to its degrees of freedom (CMIN/DF) = 1.479, and Adjusted Goodness of Fit Index (AGFI) = 0.804.

III. UX IMPROVEMENT OF AI-BASED DEFENSE SYSTEM

A. ORIGINAL SYSTEM

This section describes the upgradation process of the UX of an AI-based defense system. The original interface is illustrated in Figure 4. The main functions of the system are change detection [57], i.e., detection of terrain changes using a dense Siamese network [58], and target detection [59], i.e., detection of fighters using YOLOv3 [60]. The map function is used to assist the main function, enlarge or reduce the detected result, and monitor locations. When the change detection module detects changes in the terrain, two images of the same location at different times are displayed. By contrast, target detection relies on a single image as it merely detects the presence of airplanes at particular

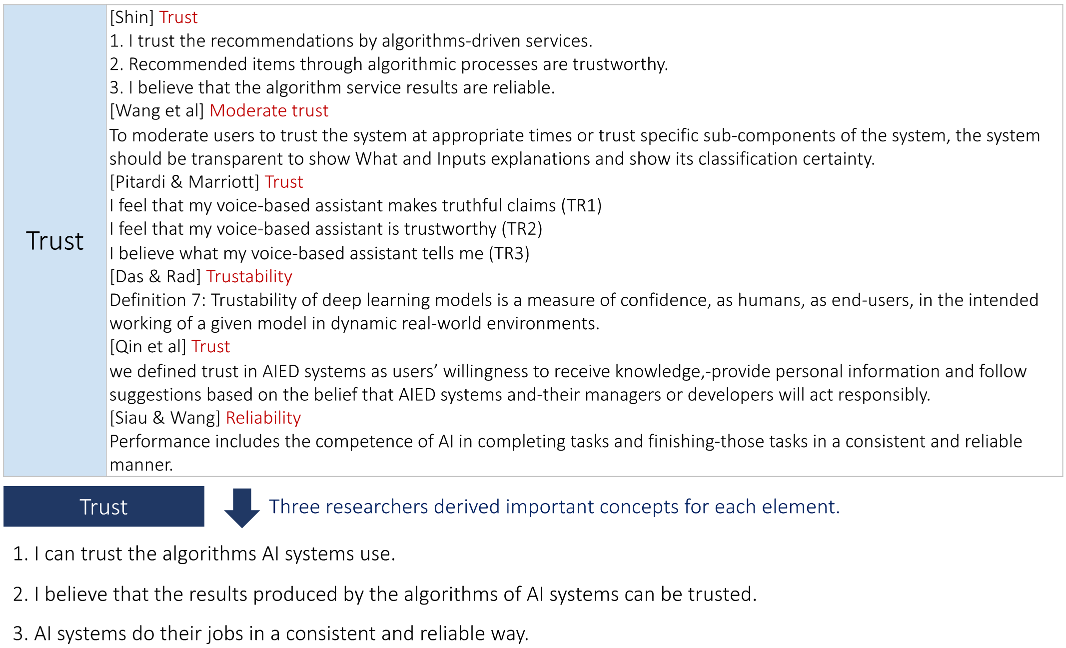


FIGURE 2. Example of AI UX element definition.

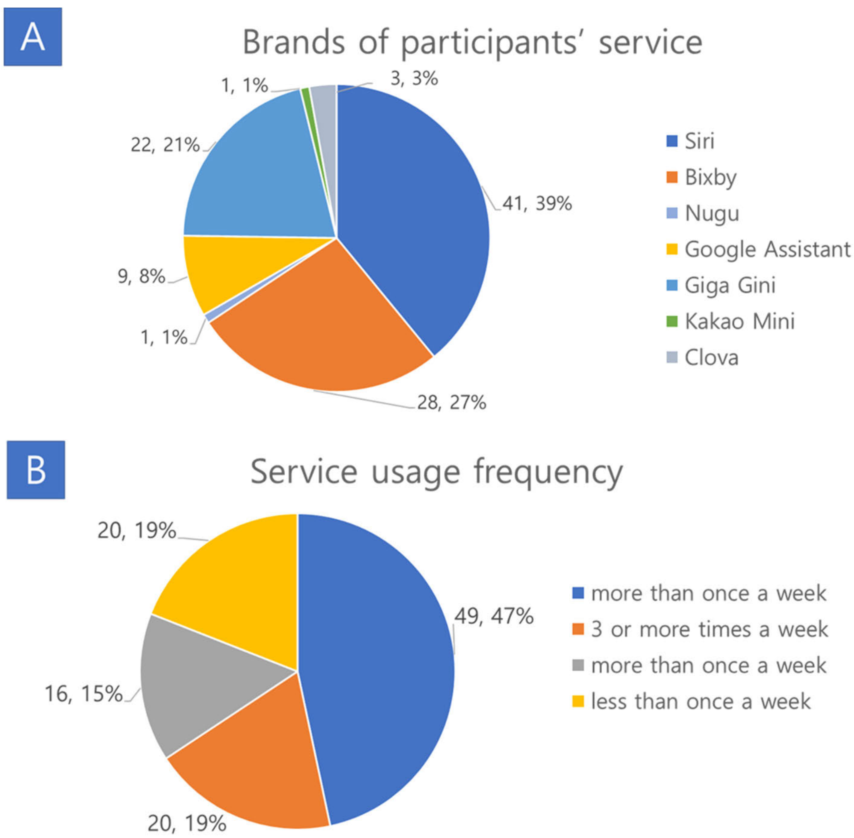


FIGURE 3. Participants' usage data.

locations. The features of the map function include search map, which displays the coordinates of the image on the map; change detectable, which displays the list of images and possible change detection images; and two overlay features, which overlay images on satellite/general maps. These four features are included in the analysis.

TABLE 1. AI UX evaluation questionnaire and factor loading.

Factor	Questionnaire	Factor loading	α
Satisfaction (Shin [19]; Amershi et al. [18]; Siau and Wang, 2018 [63]; Arnold et al. [24]; Bitkina et al. [15]; Gunning [7]; Gunning and Aha [25]; Kulesza et al. [12]; van der Waa et al. [26]; Wang et al. [17]; Chromik and Schuessler [27]; Kulesza et al. [28]; Guo [29]; Ribeiro et al. [30]; Daronnat et al. [31]; Došilović et al. [32]; Zhanget al., 2020 [52])	1. Overall, the algorithm satisfies my expectations.	0.753	0.923
	2. I feel like I am being offered the right result for my needs.	0.743	
	3. In general, I am satisfied with the results provided by the algorithm.	0.706	
	4. I think the results of AI systems are generally accurate.	0.675	
Safety (Shin [19]; Smuha [39]; Smuha [39]; Toreini et al. [40]; Siau and Wiau, 2018; Arnold et al. [24]; Pieters [41]; Fjeld et al. [42]; Barredo Arrieta et al. [43]; Kulesza et al. [12]; Rossi [44]; Robinson [45]; Shneiderman [46]; Gerlings et al. [33]; Qin et al. [34]; Royal Society [47]; Pitardi and Marriott [35]; Carvalho et al. [36])	1. I believe that AI systems are managing data safely.	0.914	0.918
	2. I believe that personal information will not be misused by the AI system.	0.841	
	3. I believe that the AI algorithm interacts with safe algorithms.	0.763	
Controllability (Amershi et al. [18]; Arnold et al. [24]; Fjeld et al. [42]; Kulesza et al. [12]; Rossi [44])	1. I can directly manipulate the detailed functions of the AI system.	0.876	0.876
	2. The functions of the AI system can be reviewed and controlled by the user.	0.751	
	3. I experienced a feeling of direct manipulation while using the AI system.	0.716	
Trust (Shin [19]; Toreini et al. [40]; Siau and Wang, 2018 [63]; Arnold et al. [24]; Pieters [41]; Bitkina et al. [15]; Glikson and Woolley [48]; Gunning [7]; Gunning and Aha [25]; Davis et al. [49]; Cheng et al. [14]; Wang et al. [17]; Rossi [44]; Das and Rad, 2020 [62]; Robinson [45]; Hussain et al. [50]; Chromik and Schuessler [27]; Kulesza et al. [28]; Guo [29]; Ribeiro et al. [30]; Bussone et al. [51]; Daronnat et al. [31]; Shneiderman [46]; Zhang et al. [52])	1. I believe that the results produced by the algorithms of AI systems can be trusted.	0.789	0.925
	2. I can trust the algorithms AI systems use.	0.736	
	3. AI systems do their jobs in a consistent and reliable way.	0.597	
Causality (Shin [19]; van der Waa et al. [26]; Wang et al. [17]; Daronnat et al. [31])	1. While using the AI system, no explanation is needed to understand the context.	0.798	0.845
	2. I did not need any support to understand the AI system's explanation.	0.762	
Fairness (Shin [19]; Smuha [39]; Smuha [39]; Toreini et al. [40]; Arnold et al. [24]; Fjeld et al. [42]; Barredo Arrieta et al. [43]; Adadi and Berrada [53]; Rossi [44]; Das and Rad, 2020[62]; Abdul et al. [54]; Gerlings et al. [33]; Qin et al. [34])	1. AI systems do their work in an unbiased and impartial due-process manner.	0.790	0.821
	2. AI systems do their jobs fairly, without discrimination or favoritism.	0.774	
Efficiency (Amershi et al. [18]; Chromik and Schuessler [27]; Qin et al. [34]; Pitardi and Marriott [35]; Carvalho et al. [36]; Nunes and Jannach [37])	1. AI systems help process information quickly and accurately.	0.702	0.803
	2. AI systems can make decisions faster than manual methods.	0.686	
Accountability (Shin [19]; Smuha [39]; Jacovi et al. [55]; Fjeld et al. [42]; Barredo Arrieta et al. [43]; Kulesza et al. [12]; Adadi and Berrada [53]; Rossi [44]; Kulesza et al. [28]; Abdul et al. [54]; Gerlings et al. [33]; Dignum [56])	1. AI systems are designed to review whether algorithms are functioning properly.	0.718	0.727
	2. Algorithms in AI systems have the ability to modify the system as a whole through specific actions.	0.717	
Explainability (Shin [19]; Amershi et al. [18]; Toreini et al. [40]; Siau and Wang, 2018 [63]; Arnold et al. [24]; Pieters [41]; Gunning [7]; Gunning and Aha [25]; Fjeld et al. [42]; Barredo Arrieta et al. [43]; Kulesza et al. [12]; van der Waa et al. [26]; Cheng et al. [14]; Adadi and Berrada [53]; Wang et al. [17]; Rossi [44]; Hussain et al. [50]; Guo [29]; Abdul et al. [54]; Ribeiro et al. [30]; Bussone et al. [51]; Došilović et al. [32])	1. One can easily understand the behavior of the algorithms of AI systems.	0.615	0.799
	2. I think the results of the AI system are interpretable.	0.575	

B. IDEATION

In the second step, identification is performed for each factor of the AI UX evaluation questionnaire (Table 1). Three researchers are involved in this phase. Potential inconveniences and improvements corresponding to each factor are identified and shared using sticker memos for

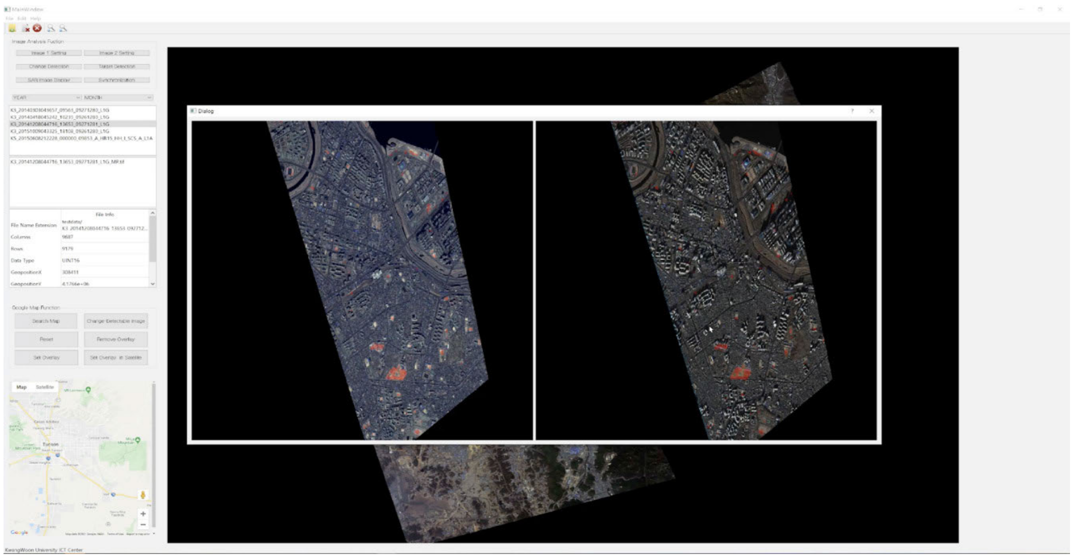


FIGURE 4. Interface of original AI defense system.

5 min. On this basis, ideation is performed for overall upgradation, with representative ideas including user guide functions, such as the help function (Figure 5(a)), and disabled representation (Figure 5(b)). Moreover, the provision of additional data to help specify and explain the reasons behind judgments of change and target detection is also decided upon (Figure 5(c)). Other examples are listed in Table 2.

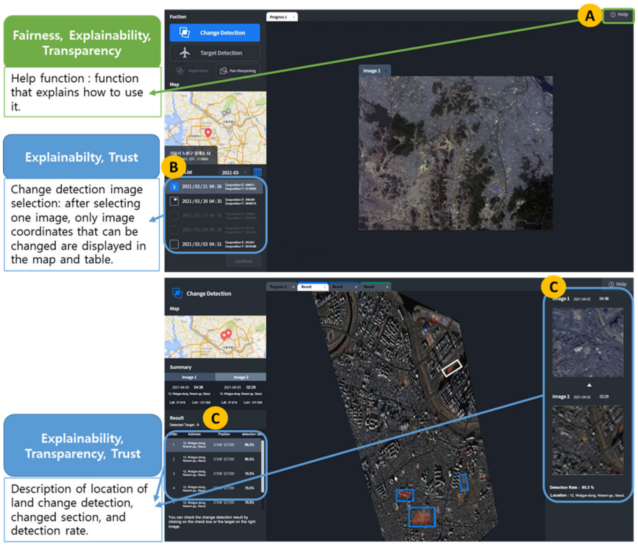


FIGURE 5. Example of ideation.

C. PROTOTYPING

In the third step, the derived ideas are visualized using Adobe XD, a design collaboration tool that improves aesthetic elements that are difficult to derive during the AI ideation stage. Examples of improved aesthetic elements include 1) changing the screen arrangement depending on the order of system use, 2) changing the text size and font, 3) changing the interface color considering visibility, and 4) securing unity by designating representative colors for change detection and target detection. The appearance of the system after improvement is depicted in Figure 6.

IV. USABILITY TEST FOR THE ADVANCED AI SYSTEM

A. PARTICIPANTS

To evaluate the usability of the advanced AI defense system described in Section III, a study is conducted on 20 participants (Mean Age = 24.05 years, SD of age = 3.67 years), consisting of 15 males and five females. The participants are recruited from the university community and are rewarded with mobile gift certificates worth approximately $ 20 each. To minimize differences in prior knowledge, participants with no prior knowledge of the national defense system were recruited. This study design was approved by the Institutional Review Board of Kwangwoon University (7001546-20210701-HR(SB)-007-04).

B. PROCEDURE

The purpose of the experiment, the methodology of using a simple system and prototype, and the experimental procedure are described to the participants. The participants are instructed to use the systems before and after upgradation, as described in Section III, on a 32-inch UHD monitor (S32A700NWK), and surveys and interviews are conducted after using each system. A balanced Latin square design is used to determine the order of system usage to avoid prediction and learning effects. The two experimental scenarios consist of the same tasks in the system before and after upgradation (Table 3). The usability evaluation items are taken to be the AI UX evaluation indicators developed in Section II, and an 11-point scale ranging from 0–10 is used to record responses, with zero representing 'not very much', and 10 representing 'very much'. The duration of the experiment

TABLE 2. Ideation results.

Factor	Ideas
Explainability	1. Change Detection: Provide detection location, range, detection rate 2. Target Detection: Provide aircraft location, number, and detection rate 3. Help function 4. Guide function during first use 5. Disable Disabled Features 6. Display only the selected image and a list of images obtained using change detection 7. Target Detection: Providing training data similar to detection results 8. Provide a prior description of the algorithm (data set, learning rate, number of epochs, etc.)
Transparency	1. Target Detection: Provide training data similar to detection results 2. Provide an advance description of the algorithm 3. Change Detection: Provide detection location, range, detection rate 4. Target Detection: Provide aircraft location, number, and detection rate 5. Help function 6. Guide function for first use

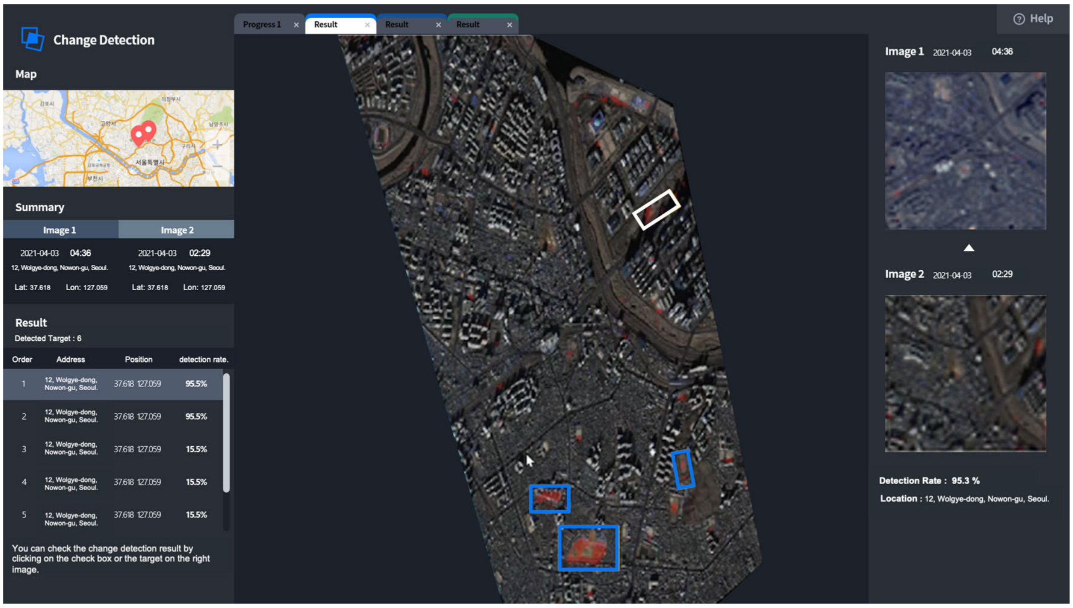


FIGURE 6. System interface after upgradation.

is approximately 1 h, and the participants are allowed to rest or discontinue whenever they wish.

C. RESULTS

The results are analyzed using repeated-measures Analysis of Variance (ANOVA). The independent variable is taken to be system type, and the nine factors defined in Table 1 are used as the dependent variables. Analysis is performed using the R software (R Development Core Team, 2006), and the results are presented in Table 4.

The usability evaluation results are depicted in Figure 7. The values of the parameters before and after upgradation are as follows: satisfaction = 5.28 (before, SD = 2.24) and 8.25 (after, SD = 1.81); safety = 5.50 (before, SD = 2.59) and 6.87 (after, SD = 1.96); controllability = 5.77 (before, SD = 1.79) and 8.07 (after, SD = 1.30); trust = 5.42 (before, SD = 2.11) and 7.90 (after, SD = 1.75); causality = 3.48 (before, SD = 2.06) and 6.88 (after, SD = 2.43); fairness = 7.35 (before, SD = 2.39) and 8.15 (after, SD = 1.88); efficiency = 6.60 (before, SD = 2.17) and 8.68 (after, SD = 1.21); accountability = 4.73 (before, SD = 2.46) and 7.23 (after, SD = 2.11); and explainability = 5.25 (before, SD = 1.93) and 8.05 (after, SD = 1.73). Thus, all AI UX factors except fairness are observed to improve after upgradation, and the differences are statistically significant.

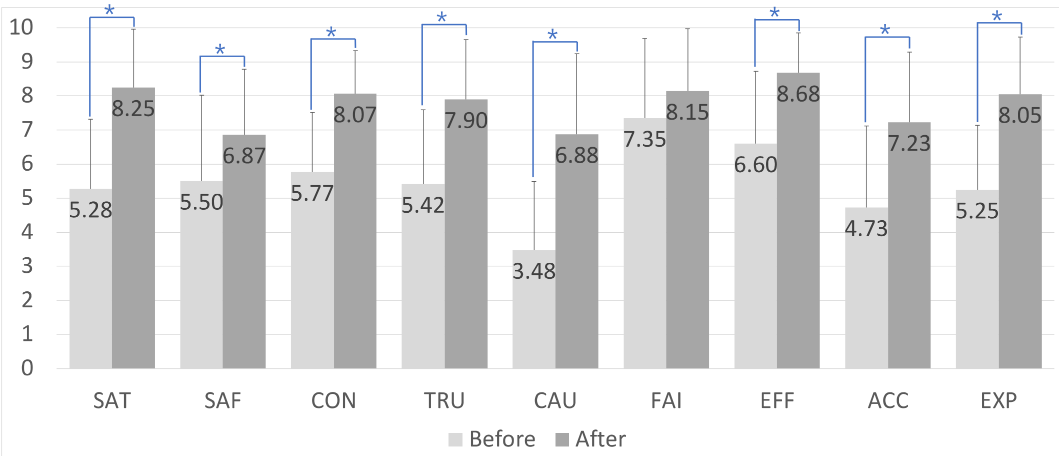
The results of participant interviews following system improvements are presented in Table 5. 90 % of the participants are observed to respond positively to the proposed upgradation process. This can be attributed to the explanation of how to use a friendly UI and additional explanations of the obtained results being provided to them. 40 % of the participants are observed to record some negative thoughts on the system. Most of these comments involve the limitations of the prototype, which could not be implemented in all cases. 45 % of the participants are observed to state their desire for additional features. In particular, additional explanation

TABLE 3. Examples of usability test scenarios.

	Before	After
Change Detection	1. Please filter the date to March 2014. 2. After selecting Image 1, press "Image 1 setting" to set it. 3. Filter the date to December 2014. 4. After selecting Image 2, press the "Image 2 setting" button above to set the image. 5. Click the "Image registration" button to align the overlapping images. 6. Check the result by clicking the "Change detection" button.	1. Click the "Change Detection" button on the home page. 2. Click the calendar icon in the 'Image List'. 3. Please filter the date settings to March 2021. 4. Click the checkbox of Image 1 in the 'Image List' to set it. 5. Click the checkbox of Image 2 in the 'Image List' to set it. 5. Align the overlapping images by clicking the "Registration" button activated above. 6. Click the "Confirm" button below to check the detected results. 7. Click the first item in 'Result' on the left to check the detailed results of Change Detection or click the blue targets in the image on the right to check the Change Detection results.

TABLE 4. Data analysis results.

	F	p		F	p		F	p
Satisfaction	26.304	0.000	**Trust**	17.829	0.000	**Efficiency**	24.012	0.000
Safety	13.597	0.002	**Causality**	40.750	0.000	**Accountability**	11.995	0.003
Controllability	33.392	0.000	**Fairness**	2.854	0.107	**Explainability**	26.435	0.000



SAT (satisfaction); SAF (safety); CON (controllability); TRU (trust); CAU (causality); FAI (fairness); EFF (efficiency); ACC (accountability); EXP (explainability). Items marked with * indicate scores with statistically significant differences before and after enhancement.

FIGURE 7. Usability evaluation results.

elements other than the developed technology, such as additional detection functions and real-time imaging, are mentioned.

V. DISCUSSION

A. DEVELOPMENT OF AI UX EVALUATION QUESTIONNAIRE

In this study, the initial items are created based on a review of existing literature to develop an AI UX evaluation tool. The questionnaire is administered to users of AI speakers to verify the appropriateness of the developed evaluation tool. Consequently, an evaluation tool is developed, comprising nine factors—satisfaction, safety, controllability, trust, causality, fairness, efficiency, accountability, and explainability— and 23 sub-items. As the aim of the developed evaluation tool is to synthesize UX factors impacting the overall AI system, overall AI UX factors are collected during the literature review, rather than being limited to a specific AI

TABLE 5. Interview results (regarding the upgraded system).

Question	Representation
Benefit	- Additional descriptions of program functions and provision of tutorials and guides - Friendly UI (Icons, text size is optimal, can be used intuitively, friendly design) - Additional description of detected results (can be viewed by comparing detected results, numerical explanations, such as detection rate, have been added, detailed detected information can be checked)
Desired features	- Limitations of prototype implementation (e.g., limited scenario configuration, scenario order is determined) - Features are described in English
Would like to add	- Additional descriptions (additional technical description to help, provide additional information, additional information such as image resolution, shooting equipment, photographer, etc.) - Additional features (detection of features other than aircraft and land changes, real-time images)

system. Therefore, the tool exhibits high generalizability and applicability as it configures general AI characteristics and does not reflect the individual characteristics of any system. The developed tool includes causality and explainability factors, which can be used to supplement the black box characteristics of artificial intelligence that users face in the decision-making process. Moreover, the AI UX evaluation tool is meaningful because of high usability—efficiency and controllability are added compared to a previous study [19]. Usability refers to the degree to which users are satisfied with the efficiency and effectiveness of the development of a user interface [61]. The sub-items corresponding to usability are diverse. Efficiency, which represents the execution quality of a specific task in terms of time and energy, as well as directness, which represents whether a user can directly control the user interface of a product/service, are also important—these are also included as subcategories [22].

B. APPLICATION TO AI SYSTEM DESIGN

In this study, the UX of an AI-based defense system is improved. The advanced stage comprises three phases: 1) task hierarchy analysis and improvement of system understanding, 2) ideation based on the elements of the AI UX factors, and 3) aesthetic design improvement and prototyping. Although several previous studies have proposed such factors, the application of these factors to systems has not been investigated adequately. The process proposed herein improves task performance and aesthetic interfaces using the aforementioned three-step advancement process. In addition, usability evaluation confirms that all prototypes exhibit statistically significant UX improvements after upgradation, except in terms of fairness. Fairness consists of items related to algorithmic bias and is not judged as an item that can cause a significant difference to UX in the system considered in this study. Analysis of the interview results reveals that positive user opinions correspond to a mixture of AI-specific items and existing UX elements. Explaining program functions in an easy-to-understand manner to human users and providing tutorials and guides have been emphasized as important, even in traditional UX research on non-AI systems. However, to help users without adequate background knowledge to understand the explanation of the operational mechanism and function of an AI system with high technical difficulty, this item requires both a technical approach and consideration from the user's perspective. Opinions grouped under friendly UI correspond to easily intelligible and highly visible interface design, with interface structures that fit the order of use. In the additional description of the detected results, both a technical approach, which displays data confirming the basis of the AI system's judgment, and a considered approach, which presents the detected results intuitively and with high visibility, are important. Finally, content related to technology and data is included as desired features to be added later. As such, most of the content suggested in the interview results (Table 5) was related to traditional UX research. For example, providing instructions and explanations for the system and its functions are not limited to defense systems. Therefore, in order to improve the UX of AI systems, it is important to consider the characteristics of artificial intelligence technology, but also consider the main considerations in traditional UX.

C. LIMITATION AND FUTURE WORK

In this study, we have developed the tool for evaluating AI UX and proposed a three-step process for enhancing UX. Furthermore, we applied this evaluation tool for conducting usability assessments. However, there are still several limitations to consider. First, a relatively small number of participants were used for metric validation. While we utilized data from 105 participants in this study for validating our questionnaire, obtaining a larger sample size for additional validation could yield more comprehensive results. Secondly, in this study, an AI based defense system was upgraded using the UX evaluation factors (Table 1). If AI systems of various characteristics are additionally considered in the future study, it is believed that the three-step advancement process proposed in this study can be generalized. Thirdly, people in their 20s participated in the user survey on AI speakers, and non-expert people with no knowledge of national defense and AI technologies participated in the usability test. Therefore, there is a need to consider diverse types of participants in future studies.

VI. CONCLUSION

Owing to the versatility and excellent performance of AI, its scope of applications has expanded in recent years. In turn, improvement of AI UX has become a popular topic of research. Most studies on this topic attempt to resolve the black-box characteristics of AI, with a focus on technological innovation. However, both technological innovations and traditional UX aspects must be considered to advance AI UX holistically. In particular, it is important to consider the characteristics of AI by expanding the traditional UX. To this end, stakeholders with varied background knowledge must be consulted during the AI system planning and development stage. In this study, an extensive literature review is conducted to identify UX elements relevant to AI UX at present, and a UX evaluation tool is developed on this basis. Subsequently, the UI of an AI defense system is upgraded via a three-step process based on the developed AI UX factors. The appropriateness of the evaluation tool is then verified—the usability evaluation results reveal that the upgraded system exhibits a more positive UX compared to the original one. This work is an initial study conducted on AI UX and is relatively limited. However, we intend to apply the tool to various AI services and systems in future works, and subject it to additional verification. Based on these improvements, we expect the proposed methodology to be generalized and used for all types of AI systems. In particular, AI UX is a relatively new topic of interest—thus, its true importance to AI design may be greater than is apparent currently. Research on AI UX should be conducted in tandem with research on AI systems, and the proposed methodology is expected to be useful in processing basic data to define the UX elements of unified AI. The results of this study are expected to be used not only as a UX evaluation tool for AI systems and services but also to evaluate the overall process of UX advancement of AI systems.

CONFLICT OF INTEREST

The authors declare that this research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

Sunyoung Park: Original draft writing, data analysis; Hyun K. Kim: Conceptualization, writing–review and editing; Jaehyun Park: Writing–review and editing; Yuryeon Lee: Data analysis, prototype design. All authors contributed to the article and approved the submitted version.

DATA AVAILABILITY STATEMENT

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

REFERENCES

[1] P. Stone, R. Brooks, E. Brynjolfsson, R. Calo, O. Etzioni, G. Hager, J. Hirschberg, S. Kalyanakrishnan, E. Kamar, S. Kraus, K. Leyton-Brown, D. Parkes, W. Press, A. Saxenian, J. Shah, M. Tambe, and A. Teller, "Artificial intelligence and life in 2030: The one hundred year study on artificial intelligence," 2022, *arXiv:2211.06318*.

[2] J. McCarthy, M. L. Minsky, N. Rochester, and C. E. Shannon, "A proposal for the Dartmouth summer research project on artificial intelligence," *AI Mag.*, vol. 27, no. 4, p. 12, 1955, doi: 10.1609/aimag.v27i4.1904.

[3] J. McCarthy, *What is Artificial Intelligence*. Dordrecht, The Netherlands: Springer, 2007.

[4] A. M. Turing, *Computing Machinery and Intelligence*. Berlin, Germany: Springer, 2009.

[5] S. J. Russell, *Artificial Intelligence a Modern Approach*. London, U.K.: Pearson, 2010.

[6] Y. Mintz and R. Brodie, "Introduction to artificial intelligence in medicine," *Minimally Invasive Therapy Allied Technol.*, vol. 28, no. 2, pp. 73–81, Mar. 2019, doi: 10.1080/13645706.2019.1575882.

[7] D. Gunning, "Explainable artificial intelligence (XAI)," *Defense Adv. Res. Projects Agency*, vol. 2, no. 2, pp. 1–35, 2017.

[8] W. Xu, "Toward human-centered AI: A perspective from human-computer interaction," *Interactions*, vol. 26, no. 4, pp. 42–46, Jun. 2019, doi: 10.1145/3328485.

[9] L. Alben, "Quality of experience: Defining the criteria for effective interaction design," *Interactions*, vol. 3, no. 3, pp. 11–15, May 1996, doi: 10.1145/235008.235010.

[10] L. Arhippainen and M. Tähti, "Empirical evaluation of user experience in two adaptive mobile application prototypes," in *Proc. 2nd Int. Conf. Mobile Ubiquitous Multimedia*, 2003, pp. 27–34.

[11] L. Luther, V. Tiberius, and A. Brem, "User experience (UX) in business, management, and psychology: A bibliometric mapping of the current state of research," *Multimodal Technol. Interact.*, vol. 4, no. 2, p. 18, May 2020, doi: 10.3390/mti4020018.

[12] T. Kulesza, M. Burnett, W.-K. Wong, and S. Stumpf, "Principles of explanatory debugging to personalize interactive machine learning," in *Proc. 20th Int. Conf. Intell. User Interfaces*, Mar. 2015, pp. 126–137, doi: 10.1145/2678025.2701399.

[13] P. Sunyoung, K. K. Hyun, S. Yeji, B. Seongjin, K. Jungyoon, P. Jangwoon, and P. Jaehyun, "Impact of expectation and performance on the user experience of AI systems," *ICIC Exp. Lett. B*, vol. 13, no. 1, pp. 73–80, 2022, doi: 10.24507/icicelb.13.01.73.

[14] H.-F. Cheng, R. Wang, Z. Zhang, F. O'connell, T. Gray, F. M. Harper, and H. Zhu, "Explaining decision-making algorithms through UI: Strategies to help non-expert stakeholders," in *Proc. Chi Conf. Human Factors Comput. Syst.*, 2019, pp. 1–12, doi: 10.1145/3290605.3300789.

[15] O. V. Bitkina, H. Jeong, B. C. Lee, J. Park, J. Park, and H. K. Kim, "Perceived trust in artificial intelligence technologies: A preliminary study," *Hum. Factors Ergonom. Manuf. Service Industries*, vol. 30, no. 4, pp. 282–290, Jul. 2020, doi: 10.1002/hfm.20839.

[16] O. Vl. Bitkina, J. Kim, J. Park, J. Park, and H. K. Kim, "User stress in artificial intelligence: Modeling in case of system failure," *IEEE Access*, vol. 9, pp. 137430–137443, 2021, doi: 10.1109/ACCESS.2021.3117120.

[17] D. Wang, Q. Yang, A. Abdul, and B. Y. Lim, "Designing theory-driven user-centric explainable AI," in *Proc. CHI Conf. Human Factors Comput. Syst.*, May 2019, pp. 1–15, doi: 10.1145/3290605.3300831.

[18] S. Amershi, D. Weld, M. Vorvoreanu, A. Fourney, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, J. Teevan, R. Kikin-Gil, and E. Horvitz, "Guidelines for human-AI interaction," in *Proc. CHI Conf. Human Factors Comput. Syst.*, May 2019, pp. 1–13, doi: 10.1145/3290605.3300233.

[19] D. Shin, "The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI," *Int. J. Hum.-Comput. Stud.*, vol. 146, Feb. 2021, Art. no. 102551, doi: 10.1016/j.ijhcs.2020.102551.

[20] K. Kieslich, B. Keller, and C. Starke, "Artificial intelligence ethics by design. Evaluating public perception on the importance of ethical design principles of artificial intelligence," *Big Data Soc.*, vol. 9, no. 1, Jan. 2022, Art. no. 205395172210929, doi: 10.1177/20539517221092956.

[21] Q. Zheng, Y. Tang, Y. Liu, W. Liu, and Y. Huang, "UX research on conversational human-AI interaction: A literature review of the ACM digital library," in *Proc. CHI Conf. Human Factors Comput. Syst.*, Apr. 2022, pp. 1–24, doi: 10.1145/3491102.3501855.

[22] J. Park, S. H. Han, H. K. Kim, Y. Cho, and W. Park, "Developing elements of user experience for mobile phones and services: Survey, interview, and observation approaches," *Hum. Factors Ergonom. Manuf. Service Industries*, vol. 23, no. 4, pp. 279–293, Jul. 2013, doi: 10.1002/hfm.20316.

[23] E. L.-C. Law, V. Roto, M. Hassenzahl, A. P. O. S. Vermeeren, and J. Kort, "Understanding, scoping and defining user experience: A survey approach," in *Proc. SIGCHI Conf. Human Factors Comput. Syst.*, Apr. 2009, pp. 719–728, doi: 10.1145/1518701.1518813.

[24] M. Arnold, R. K. E. Bellamy, M. Hind, S. Houde, S. Mehta, A. Mojsilovic, R. Nair, K. N. Ramamurthy, A. Olteanu, D. Piorkowski, D. Reimer, J. Richards, J. Tsay, and K. R. Varshney, "FactSheets: Increasing trust in AI services through supplier's declarations of conformity," *IBM J. Res. Develop.*, vol. 63, pp. 1–13, Jul. 2019, doi: 10.1147/JRD.2019.2942288.

[25] D. Gunning and D. W. Aha, "DARPA's explainable artificial intelligence program," *AI Mag.*, vol. 40, no. 2, pp. 44–58, Jun. 2019, doi: 10.1609/aimag.v40i2.2850.

[26] J. van der Waa, E. Nieuwburg, A. Cremers, and M. Neerincx, "Evaluating XAI: A comparison of rule-based and example-based explanations," *Artif. Intell.*, vol. 291, Feb. 2021, Art. no. 103404, doi: 10.1016/j.artint.2020.103404.

[27] M. Chromik and M. Schuessler, "A taxonomy for human subject evaluation of black-box explanations in XAI," in *Proc. IUI Workshop Explainable Smart Syst. Algorithmic Transparency Emerg. Technol. (ExSS-ATEC)*, Cagliari, Italy, 2020.

[28] T. Kulesza, S. Stumpf, M. Burnett, S. Yang, I. Kwan, and W.-K. Wong, "Too much, too little, or just right? Ways explanations impact end users' mental models," in *Proc. IEEE Symp. Vis. Lang. Human Centric Comput.*, Sep. 2013, pp. 3–10, doi: 10.1109/VLHCC.2013.6645235.

[29] W. Guo, "Explainable artificial intelligence for 6G: Improving trust between human and machine," *IEEE Commun. Mag.*, vol. 58, no. 6, pp. 39–45, Jun. 2020, doi: 10.1109/MCOM.001.2000050.

[30] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.

[31] S. Daronnat, L. Azzopardi, M. Halvey, and M. Dubiel, "Impact of agent reliability and predictability on trust in real time human-agent collaboration," in *Proc. 8th Int. Conf. Human-Agent Interact.*, Nov. 2020, pp. 131–139, doi: 10.1145/3406499.3415063.

[32] F. K. Dosilovic, M. Brcic, and N. Hlupic, "Explainable artificial intelligence: A survey," in *Proc. 41st Int. Conv. Inf. Commun. Technol., Electron. Microelectron. (MIPRO)*, May 2018, doi: 10.23919/MIPRO.2018.8400040.

[33] J. Gerlings, A. Shollo, and I. Constantiou, "Reviewing the need for explainable artificial intelligence (xAI)," 2020, *arXiv:2012.01007*.

[34] F. Qin, K. Li, and J. Yan, "Understanding user trust in artificial intelligence-based educational systems: Evidence from China," *Brit. J. Educ. Technol.*, vol. 51, no. 5, pp. 1693–1710, Sep. 2020, doi: 10.1111/bjet.12994.

[35] V. Pitardi and H. R. Marriott, "Alexa, she's not human but…Unveiling the drivers of consumers' trust in voice-based artificial intelligence," *Psychol. Marketing*, vol. 38, no. 4, pp. 626–642, Apr. 2021, doi: 10.1002/mar.21457.

[36] D. V. Carvalho, E. M. Pereira, and J. S. Cardoso, "Machine learning interpretability: A survey on methods and metrics," *Electronics*, vol. 8, no. 8, p. 832, Jul. 2019, doi: 10.3390/electronics8080832.

[37] I. Nunes and D. Jannach, "A systematic review and taxonomy of explanations in decision support and recommender systems," *User Model. User-Adapted Interact.*, vol. 27, nos. 3–5, pp. 393–444, Dec. 2017, doi: 10.1007/s11257-017-9195-0.

[38] Z. Ashktorab, C. Dugan, J. Johnson, Q. Pan, W. Zhang, S. Kumaravel, and M. Campbell, "Effects of communication directionality and AI agent differences in human-AI interaction," in *Proc. CHI Conf. Human Factors Comput. Syst.*, 2021, pp. 1–15.

[39] N. A. Smuha, "The EU approach to ethics guidelines for trustworthy artificial intelligence," *Comput. Law Rev. Int.*, vol. 20, no. 4, pp. 97–106, Aug. 2019, doi: 10.9785/cri-2019-200402.

[40] E. Toreini, M. Aitken, K. Coopamootoo, K. Elliott, C. G. Zelaya, and A. van Moorsel, "The relationship between trust in AI and trustworthy machine learning technologies," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2020, pp. 272–283, doi: 10.1145/3351095.3372834.

[41] W. Pieters, "Explanation and trust: What to tell the user in security and AI?" *Ethics Inf. Technol.*, vol. 13, no. 1, pp. 53–64, 2011, doi: 10.1007/s10676-010-9253-3.

[42] J. Fjeld, N. Achten, H. Hilligoss, A. Nagy, and M. Srikumar, "Principled artificial intelligence: Mapping consensus in ethical and rights-based approaches to principles for AI," *SSRN Electron. J.*, no. 2020-1, Jan. 2020, doi: 10.2139/ssrn.3518482.

[43] A. Barredo Arrieta, N. Díaz-Rodríguez, J. Del Ser, A. Bennetot, S. Tabik, A. Barbado, S. Garcia, S. Gil-Lopez, D. Molina, R. Benjamins, R. Chatila, and F. Herrera, "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Inf. Fusion*, vol. 58, pp. 82–115, Jun. 2020, doi: 10.1016/j.inffus.2019.12.012.

[44] F. Rossi, "Building trust in artificial intelligence," *J. Int. Aff.*, vol. 72, no. 1, pp. 127–134, 2018.

[45] S. C. Robinson, "Trust, transparency, and openness: How inclusion of cultural values shapes Nordic national public policy strategies for artificial intelligence (AI)," *Technol. Soc.*, vol. 63, Nov. 2020, Art. no. 101421, doi: 10.1016/j.techsoc.2020.101421.

[46] B. Shneiderman, "Human-centered artificial intelligence: Reliable, safe & trustworthy," *Int. J. Hum.-Comput. Interact.*, vol. 36, no. 6, pp. 495–504, Apr. 2020, doi: 10.1080/10447318.2020.1741118.

[47] Royal Society. (2019). *Explainable AI: The Basics. Policy Briefing*. [Online]. Available: https://royalsociety.org/-/media/policy/projects/explainable-ai/AI-and-interpretability-policy-briefing.pdf

[48] E. Glikson and A. W. Woolley, "Human trust in artificial intelligence: Review of empirical research," *Acad. Manage. Ann.*, vol. 14, no. 2, pp. 627–660, Jul. 2020, doi: 10.5465/annals.2018.0057.

[49] B. Davis, M. Glenski, W. Sealy, and D. Arendt, "Measure utility, gain trust: Practical advice for XAI researchers," in *Proc. IEEE Workshop TRust EXpertise Vis. Analytics (TREX)*, Oct. 2020, pp. 1–8, doi: 10.1109/TREX51495.2020.00005.

[50] F. Hussain, R. Hussain, and E. Hossain, "Explainable artificial intelligence (XAI): An engineering perspective," 2021, *arXiv:2101.03613*.

[51] A. Bussone, S. Stumpf, and D. O'Sullivan, "The role of explanations on trust and reliance in clinical decision support systems," in *Proc. Int. Conf. Healthcare Informat.*, Oct. 2015, pp. 160–169, doi: 10.1109/ICHI.2015.26.

[52] Y. Zhang, Q. V. Liao, and R. K. E. Bellamy, "Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making," in *Proc. Conf. Fairness, Accountability, Transparency*, Jan. 2020, pp. 295–305, doi: 10.1145/3351095.3372852.
[53] A. Adadi and M. Berrada, "Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)," *IEEE Access*, vol. 6, pp. 52138–52160, 2018, doi: 10.1109/ACCESS.2018.2870052.
[54] A. Abdul, J. Vermeulen, D. Wang, B. Y. Lim, and M. Kankanhalli, "Trends and trajectories for explainable, accountable and intelligible systems: An HCI research agenda," in *Proc. CHI Conf. Human Factors Comput. Syst.*, Apr. 2018, pp. 1–18, doi: 10.1145/3173574.3174156.
[55] A. Jacovi, A. Marasović, T. Miller, and Y. Goldberg, "Formalizing trust in artificial intelligence: Prerequisites, causes and goals of human trust in AI," in *Proc. ACM Conf. Fairness, Accountability, Transparency*, Mar. 2021, pp. 624–635, doi: 10.1145/3442188.3445923.
[56] V. Dignum, "Responsible artificial intelligence: Designing AI for human values," *ITU J. ICT Discov.*, vol. 1, pp. 1–8, Sep. 2017.
[57] D. Lu, P. Mausel, E. Brondízio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, Jun. 2004, doi: 10.1080/0143116031000139863.
[58] M. H. Abdelpakey, M. S. Shehata, and M. M. Mohamed, "DensSiam: End-to-end densely-siamese network with self-attention model for object tracking," in *Proc. 13th Int. Symp.*, Las Vegas, NV, USA, Nov. 2018, pp. 463–473, doi: 10.1007/978-3-030-03801-4_41.
[59] W. Li, S. Xiang, H. Wang, and C. Pan, "Robust airplane detection in satellite images," in *Proc. 18th IEEE Int. Conf. Image Process.*, Sep. 2011, pp. 2821–2824, doi: 10.1109/ICIP.2011.6116259.
[60] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.
[61] D. Hix and H. R. Hartson, *Developing User Interfaces: Ensuring Usability Through Product and Process*. Hoboken, NJ, USA: Wiley, 1993.
[62] A. Das and P. Rad, "Opportunities and challenges in explainable artificial intelligence (XAI): A survey," 2020, *arXiv:2006.11371*.
[63] K. Siau and W. Wang, "Building trust in artificial intelligence, machine learning, and robotics," *Cutter Bus. Technol. J.*, vol. 31, no. 2, pp. 47–53, 2018.
[64] S. B. Reeder, J. Jensen, and R. Ball, "Evaluating explainable AI (XAI) in terms of user gender and educational background," in *Artificial Intelligence in HCI* (Lecture Notes in Computer Science). Cham, Switzerland: Springer, Jan. 2023, pp. 286–304, doi: 10.1007/978-3-031-35891-3_18.
[65] J. Aechtner, L. Cabrera, D. Katwal, P. Onghena, D. P. Valenzuela, and A. Wilbik, "Comparing user perception of explanations developed with XAI methods," in *Proc. IEEE Int. Conf. Fuzzy Syst. (FUZZ-IEEE)*, Jul. 2022, pp. 1–7, doi: 10.1109/fuzz-ieee55066.2022.9882743.
[66] R. R. Hoffman, S. T. Mueller, G. Klein, and J. Litman, "Measures for explainable AI: Explanation goodness, user satisfaction, mental models, curiosity, trust, and human-AI performance," *Frontiers Comput. Sci.*, vol. 5, Feb. 2023, Art. no. 1096257, doi: 10.3389/fcomp.2023.1096257.
[67] T. A. Bach, A. Khan, H. Hallock, G. Beltrão, and S. Sousa, "A systematic literature review of user trust in AI-enabled systems: An HCI perspective," *Int. J. Hum.-Comput. Interact.*, pp. 1–16, Nov. 2022, doi: 10.1080/10447318.2022.2138826.

SUNYOUNG PARK received the B.S. degree from the School of Software, Kwangwoon University, and the M.Sc. degree from the Department of Artificial Intelligence Convergence, Kwangwoon University. She is currently pursuing the Ph.D. degree with the Department of Applied Artificial Intelligence, Sungkyunkwan University. Her research interests include human-centered AI and data analytics.

HYUN K. KIM received the B.S. and Ph.D. degrees from the Department of Industrial and Management Engineering, Pohang University of Science and Technology (POSTECH). She is currently an Associate Professor with the School of Information Convergence, Kwangwoon University. Her research interests include human–AI interaction, accessibility, and Metaverse user experience.

JAEHYUN PARK received the B.S. and Ph.D. degrees in industrial and management engineering from the Pohang University of Science and Technology (POSTECH). He is currently an Associate Professor with the Department of Industrial and Management Engineering, Incheon National University. His research interests include human–computer interaction, user experience, and user value.

YURYEON LEE received the B.S. degree from the School of Information Convergence, Kwangwoon University, and the M.Sc. degree from the Department of Artificial Intelligence Convergence, Kwangwoon University. She is currently a Researcher with KT Corporation. Her research interests include usability and accessibility, UX research, and VR/AR.